44. Homological Algebra

1. Because the sequence is exact, the image of the map $0 \to A$, which must be 0, is the kernel of $f : A \to B$. Because all these maps are homomorphism, this means that f is one to one. The kernel of $B \to 0$ is certainly B, which, by exactness, must be the image $f[A]$ of A under f. Thus f is a homomorphism mapping A one to one onto B, and is thus an isomorphism.

2. **a.** The map $C \to 0$ had kernel C, which must, by exactness at C, be the image $j[B]$, that is, j maps B onto C.

b. The map $0 \to A$ has image 0, which must be the kernel of i by exactness at A. Thus the map $i : A \to B$ has kernel 0, so i maps A one to one into B.

c. Exactness at B means that the image $i[A]$ is the kernel of j. Because j is a homomorphism, we then know that $j[B] \simeq B/i[A]$. But $j[B] = C$ by Part(**a**), so we have $C \simeq B/i[A]$.

3. By exactness at B, the map i is onto B if and only if the kernel of j is B, which is true if and only if j maps B onto 0. We have shown (1) if and only if (2). By exactness at C, we see that j maps B onto 0 if and only if the kernel of the map k is 0, which holds if and only if k is a one to one map. This shows (2) if and only if (3).

4. Now exactness at C and means that h maps everything onto 0 if and only if kernel of i is 0 , which is true if and only if i is one to one. Now j maps everything onto zero if and only if the the kernel of j is D which is true, by exactness at D, if and only if the image of i is D. Thus h and j both map everything onto 0 if and only if i is one to one and maps C onto D, in other words, if and only if i is an isomorphism. We have shown (1) if and only if (2).

Now h maps everthing onto 0 if and only the kernel of h is B, which, by exactness at B, holds if and only if $g[A] = B$ that is if and only if g maps A onto B. Also exactness at E means that j maps everything onto 0 if and only if the kernel of k is 0, which is true if and only if k is one to one. Thus h and j both map everything onto 0 if and only if g is onto B and k is one to one. We have shown (1) if and only if (3).

5. (See the answer in the text.)

6. Let X be the torus complex and let Y be the subcomplex consisting of the 1-cycle a. Let P be a vertex on a and let Q be a vertex on the torus such that Q is not a vertex on a but PQ is a 1-simplex. Then $Q + C_0(Y)$ generates $Z_0(X, Y)$ and

$$\begin{aligned} \overline{\partial}_1(PQ + C_1(Y)) &= \partial_1(PQ) + C_0(Y) \\ &= (Q - P) + C_0(Y) = Q + C_0(Y) \end{aligned}$$

because $P \in C_0(Y)$. Thus our generator of $Z_0(X, Y)$ is a relative 0-boundary, so $H_0(X, Y) = 0$.

The generators of $H_1(X)$ are the cosets $a + B_1(X)$ and $b + B_1(X)$. Because $a \in Y$, we see that $b + B_1(X, Y)$ generates $H_1(X, Y)$, which is thus isomorphic to $\mathbb{Z}$.

Let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Because $\partial_2(c) = 0$, we see that

$$\overline{\partial}_2(c + C_2(Y)) = \partial_2(c) + C1(Y) = 0 + C_1(Y),$$

and $C_1(Y) = 0$ in $C_1(X, Y)$. Thus $H_2(X, Y) \simeq \mathbb{Z}$. Of course $H_n(X, Y) = 0$ for $n > 2$.

7. (See the answer in the text.)

8. Let X be the Klein bottle complex and let Y be the subcomplex consisting of the 1-cycle a. Let P be a vertex on a and let Q be a vertex on the Klein bottle such that Q is not a vertex on a but PQ is a 1-simplex. Then $Q + C_0(Y)$ generates $Z_0(X, Y)$ and

$$\begin{aligned}\overline{\partial}_1(PQ + C_1(Y)) &= \partial_1(PQ) + C_0(Y) \\ &= (Q - P) + C_0(Y) = Q + C_0(Y)\end{aligned}$$

because $P \in C_0(Y)$. Thus our generator of $Z_0(X, Y)$ is a relative 0-boundary, so $H_0(X, Y) = 0$.

The generators of $H_1(X)$ are the cosets $a + B_1(X)$ and $b + B_1(X)$. Because $a \in Y$, we see that $b + B_1(X, Y)$ generates $H_1(X, Y)$, which is thus isomorphic to $\mathbb{Z}$.

Let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Because $\partial_2(c) = 2a \in C_1(Y)$, we see that

$$\overline{\partial}_2(c + C_2(Y)) = \partial_2(c) + C_1(Y) = 2a + C_1(Y) = C_1(Y),$$

and $C_1(Y) = 0$ in $C_1(X, Y)$. Thus $H_2(X, Y) \simeq Z$. Of course $H_n(X, Y) = 0$ for $n > 2$.

9. (See the answer in the text.)

10. Let P be a vertex in Y and let R be a vertex on the annular ring such that R is not a vertex in Y but PR is a 1-simplex. Then $R + C_0(Y)$ generates $Z_0(X, Y)$ and

$$\begin{aligned}\overline{\partial}_1(PR + C_1(Y)) &= \partial_1(PR) + C_0(Y) \\ &= (R - P) + C_0(Y) = R + C_0(Y)\end{aligned}$$

because $P \in C_0(Y)$. Thus our generator of $Z_0(X, Y)$ is a relative 0-boundary, so $H_0(X, Y) = 0$.

We saw in Example 42.10 than any 1-cycle in $Z_1(X)$ could be pushed to the outer rim of the annulus. This outer rim is now part of Y, so this 1-cycle in $Z_1(X)$ becomes homologous to 0 in $Z_1(X, Y)$. However, the 1-simplex P_1Q_1 from the inner rim to the outer rim in Fig. 42.11, which was not a 1-cycle in $Z_1(X)$, now becomes a 1-cycle in $Z_1(X, Y)$, because its boundary $Q_1 - P_1$ lies in Y. In fact, for any triangulation, every sequence $R_1R_2 + R_2R_3 + \cdots + R_{m-1}R_m$ where the R_i are vertices with $R_1, R_m \in Y$ is in $Z_1(X, Y)$. However, if R_1 and R_m are both in the outer rim or both in the inner rim of the annulus, then the 1-cycle is in $B_1(X, Y)$. In terms of the triangulation in Fig. 42.11, We see that $H_1(X, Y)$ is generated by $P_1Q_1 + B_1(X, Y)$, so $H_1(X, Y) \simeq \mathbb{Z}$.

Finally, let c be the 2-chain consisting of the sum of all 2-simplexes oriented the same way. Then $\partial_2(c) \in C_1(Y)$, and $C_1(Y) = 0$ in $C_1(X, Y)$. Thus $c + C_2(Y)$ becomes a 2-cycle in $Z_2(X, Y)$, and $c + B_2(X, Y)$ is a generator of $H_2(X, Y)$ which is isomorphic to $\mathbb{Z}$. Of course $H_n(X, Y) = 0$ for $n > 2$.

11. Because $H_2(X) = 0$, the image of i_{*2} and kernel of j_{*2} are both zero, so we have exactness there.

Let a be the inner rim of the annular ring, oriented clockwise and let b be its outer rim oriented counterclockwise in Fig. 42.11. Let c be the 2-chain consisting of the sum of all 2-simplexes oriented counterclockwise. Then ∂_{*2} maps the generator $c + B_2(X, Y)$ of $H_2(X, Y)$ onto $(a + b) + B_1(Y)$ which is a nonzero element of $H_1(Y)$, so the kernel of ∂_{*2} and image of j_{*2} are both 0, and we have exactness at $H_2(X, Y)$.

We can take $-a + B_1(Y)$ and $b + B_1(Y)$ as generators of $H_1(Y)$, and i_{*1} maps these generators into $-a + B_1(X)$ and $b + B_1(X)$ respectively. Because $b + a = b - (-a)$ is in $B_1(X)$, we see that $-a + B_1(X)$ and $b + B_1(X)$ are the same homology class, which generates $H_1(X)$. Thus the image of ∂_{*2} consists of all $m(a + b) + B_1(Y)$ for $m \in \mathbb{Z}$, which are precisely the elements mapped into 0 in $H_1(X)$, because

$a+b$ generates $B_1(X)$. Thus we have exactness at $H_1(Y)$. (Note that in the identification of $H_1(Y)$ with $\mathbb{Z}\times\mathbb{Z}$ and the identification of $H_1(X)$ with $\mathbb{Z}$, the kernel of the homomorphism corresponding to i_{*1} is $\langle(1,1)\rangle$, so that $(\mathbb{Z}\times\mathbb{Z})/\langle(1,1)\rangle \simeq \mathbb{Z}$.)

Now j_{*1} maps the generator $b+B_1(X)$ of $H_1(X)$ into $0+B_1(X,Y)$ because b is in Y. We saw that $\langle b+B_1(X)\rangle = H_1(X)$ is the image under i_{*1}, so we have exactness at $H_1(X)$.

Referring to Fig. 42.11, we see that $H_1(X,Y)$ is generated by $P_1Q_1+B_1(X,Y)$, and $\partial(P_1Q_1) = Q_1-P_1$, so ∂_{*1} maps $P_1Q_1+B_1(X,Y)$ into $(Q_1-P_1)+B_0(Y)$, and $Q_1-P_1 \notin B_0(Y)$ because $P_1Q_1 \notin Y$. Thus the kernel of ∂_{*1} and the image of j_{*1} are both zero, so we have exactness at $H_1(X,Y)$.

Now$H_0(Y)$ has as generators $P_1+B_0(Y)$ and $Q_1+B_0(Y)$. These are mapped by i_{*0} into $P_1+B_0(X)$ and $Q_1+B_0(X)$ respectively. However, these are the same homology class in $H_0(X)$ because $(Q_1-P_1)\in B_0(X)$. Thus the image $m(Q_1-P_1)+B_0(Y)$ under ∂_{*1} is the kernel of i_{*0}, so we have exactness at $H_0(Y)$.

Finally, j_{*0} maps the generator $P_1+B_0(X)$ of $H_0(X)$ onto $0+B_0(X,Y)$ because $P_1\in Y$. Thus the image $mP_1+B_0(X)$ under i_{*0} is the kernel under j_{*0}, so we have exactness at $H_0(X)$.

12. Let $c\in A_k$. Then $(\overline{\partial}j_k)(c) = \overline{\partial}(j_k(c)) = \overline{\partial}(c+A'_k) = \partial(c)+A'_{k-1} = j_{k-1}(\partial(c)) = (j_{k-1}\partial)(c)$, so $\overline{\partial}j_k = j_{k-1}\partial$.

13. Let $h\in H_k(A/A')$. Let $z_1,z_2\in Z_k(A/A')$ be such that $h = z_1+B_k(A/A') = z_2+B_k(A/A')$, so that $(z_2-z_1)\in B_k(A/A')$. Let $z_1 = c_1+A'_k$ and $z_2 = c_2+A'_k$ To show that ∂_{*k} is well defined, we must show that $\partial_k(c_1)+B_{k-1}(A') = \partial_k(c_2)+B_{k-1}(A')$. Now $z_2-z_1 = c_2-c_1+A'_k$ is in $B_k(A/A')$. Consequently there is some $r\in A_{k+1}$ such that $\partial_{k+1}r = c_2-c_1+a'$ for some $a'\in A'_k$. Then $0 = \partial_k\partial_{k+1}(r) = \partial_k(c_2-c_1+a') = \partial_k(c_2)-\partial_k(c_1)+\partial_k(a')$. Now $\partial_k(a')\in B_{k-1}(A')$, so we see that $\partial_k(c_2)\in\partial_k(c_1)+B_{k-1}(A')$. This shows that ∂_{*k} is well defined.

Let $h_1,h_2\in H_k(A/A')$, and now let $z_1,z_2\in Z_k(A/A')$ be such that $h_1 = z_1+B_k(A/A')$ and $h_2 = z_2+B_k(A/A')$, so that $h_1+h_2 = (z_1+z_2)+B_k(A/A')$. Let $z_1 = c_1+A'_k$ and $z_2 = c_2+A'_k$, so that $z_1+z_2 = c_1+c_2+A'_k$. Then

$$\begin{aligned}\partial_{*k}(h_1+h_2) &= \partial_k(c_1+c_2)+B_{k-1}(A')\\ &= (\partial_k(c_1)+\partial_k(c_2))+B_{k-1}(A')\\ &= (\partial_k(c_1)+B_{k-1})(A'))+(\partial_k(c_2)+B_{k-1}(A'))\\ &= \partial_{*k}(h_1)+\partial_{*k}(h_2).\end{aligned}$$

Thus ∂_{*k} is a homomorphism.

14. Let $z'+B_k(A')$ be an element of $H_k(A')$. Then $i_{*k}(z'+B_k(A')) = z'+B_k(A)$. Now $j_{*k}(z'+B_k(A)) = (z'+A')+B_k(A/A') = 0$ because $z'\in A'$. Thus $j_{*k}i_{*k} = 0$.

Let $z+B_k(A)$ be an element of $H_k(A)$. Then $j_{*k}(z+B_k(A)) = (z+A')+B_k(A/A')$. Now $\partial_{*k}((z+A')+B_k(A/A')) = \partial_k(z)+B_{k-1}(A') = B_{k-1}(A')$ because z is a k-cycle in A_k, and $B_{k-1}(A')$ is the zero element of $H_{k-1}(A')$. Thus $\partial_{*k}j_{*k} = 0$.

Let $h = (c+A')+B_k(A/A')$ be an element of $H_k(A/A')$. We have $\partial_{*k}(h) = \partial_k(c)+B_{k-1}(A')$ where $\partial_k(c)\in A'$. Then we have $i_{*k-1}(\partial_k(c)+B_{k-1}(A')) = \partial_k(c)+B_{k-1}(A) = B_{k-1}(A)$ because $c\in A$ implies $\partial_k(c)$ is in $B_{k-1}(A)$. Now $B_{k-1}(A)$ is the zero element of $H_k(A)$, so $i_{*k-1}\partial_{*k} = 0$.

15. a. We must show that $j_{*k}i_{*k} = 0$. Let $h'\in H_k(A')$. Then $h' = z'+B_k(A')$ for some $z'\in A'_k$, and $i_{*k}(h') = z'+B_k(A)$, and $j_{*k}(z'+B_k(A)) = (z'+A'_k)+B_k(A/A')$. But $z'\in A'_k$ so $(z'+A'_k)\in B_k(A/A')$, because

$$B_k(A/A') = \{\partial_{k+1}(a_{k+1})+A'_k \mid a_{k+1}\in A_{k+1}\},$$

and taking $a_{k+1} = 0$, we get $A'_k \subseteq B_k(A/A')$. Thus $j_{*k}(i_{*k}(h')) = 0$ in $H_k(A/A')$.

b. Let $h \in H_k(A)$ and let $j_{*k}(h) = 0$ in $H_k(A/A')$. Now if $h = z + B_k(A)$, we have $j_{*k}(h) = (z + A'_k) + B_k(A/A')$, and $j_{*k}(h) = 0$ implies that $(z + A'_k) \in B_k(A/A')$. Now

$$B_k(A/A') = \{\partial_{k+1}(a_{k+1}) + A'_k \mid a_{k+1} \in A_{k+1}\}.$$

Thus $z = \partial_{k+1}(a_{k+1}) + a'_k$ for some $a_{k+1} \in A_{k+1}$ and $a'_k \in A'_k$. Because z is a k-cycle, we have

$$0 = \partial_k(z) = \partial_k(\partial_{k+1}(a_{k+1}) + \partial_k(a'_k)) = 0 + \partial_k(a'_k),$$

so a'_k is a k-cycle in A'. Therefore we see that $i_{*k}(a'_k + B_k(A')) = a'_k + B_k(A) = (z - \partial_{k+1}(a_{k+1})) + B_k(A) = z + B_k(A) = h$, because $\partial_{k+1}(a_{k+1}) \in B_k(A)$. We have shown that $h \in$ (kernel j_{*k}) is also in (image i_{*k}).

c. We must show that $\partial_{*k} j_{*k} = 0$. Let $(z+b_k(A)) \in H_k(A)$. Then $\partial_{*k}(j_{*k}(z+B_k(A))) = \partial_{*k}((z+A')+B_k(A/A')) = \partial_k(z) + B_{k-1}(A') = B_{k-1}(A')$ because z is a k-cycle so $\partial_k(z) = 0$. Because $B_{k-1}(A')$ is the 0-element of $H_{k-1}(A')$, we are done.

d. Let $h \in H_k(A/A')$ be such that $\partial_{*k}(h) = 0$, and let $h = (z + A') + B_k(A/A')$. Then $\partial_{*k}(h) = \partial_k(z) + B_{k-1}(A')$, which must be the zero element of $H_{k-1}(A')$, so $\partial_k(z) \in B_{k-1}(A')$. Let $a'_k \in A'_k$ be such that $\partial_k(a'_k) = \partial_k(z)$. Then $\partial_k(z - a'_k) = \partial_k(z) - \partial_k(a'_k) = \partial_k(z) - \partial_k(z) = 0$, so $z - a'_k$ is a k-cycle in A_k. Then we have $j_{*k}((z - a'_k) + B_k(A)) = (z - a'_k) + B_k(A/A')$. We saw in Part(**a**) that $A'_k \subseteq B_k(A/A')$ so $(z - a'_k) + B_k(A/A') = z + B_k(A/A') = h$. Thus $h \in$ (image j_{*k}).

e. We must show that $i_{*k-1}\partial_{*k} = 0$. Let $h \in H_k(A/A')$ and let $h = (z + A') + B_k(A/A')$. Now $\partial_{*k}(h) = \partial_k(z) + B_{k-1}(A')$, and $i_{*k-1}(\partial_k(z) + B_{k-1}(A')) = \partial_k(z) + B_{k-1}(A)$. But $\partial_k(z) \in B_{k-1}(A)$ and $B_{k-1}(A)$ is the zero element of $H_{k-1}(A)$. Thus $i_{*k-1}(\partial_{*k}(h)) = 0$.

f. Let $h' \in H_{k-1}(A')$ and suppose that $i_{*k-1}(h') = 0$. Let $h' = z' + B_{k-1}(A')$. Then $i_{*k-1}(h') = z'+B_{k-1}(A)$, and this is zero in $H_{k-1}(A)$ if and only if $z' \in B_{k-1}(A)$. Let $c \in A_k$ be such that $\partial_k(c) = z'$. Then $c+A'$ is a k-cycle in $H_k(A/A')$ because $\partial_k(c+A') = z'+A' = A'$. Then $\partial_{*k}((c+A')+B_k(A/A')) = z' + B_{k-1}(A') = h'$, which is what we wished to show.

16. Let z_k be a k-cycle in A_k. Using the relation in the text, we have

$$f_k(z_k) - g_k(z_k) = \partial'_{k+1}(D_k(z_k)) + D_{k-1}(\partial_k(z_k)).$$

Because z_k is a k-cycle, we have $\partial_k(z_k) = 0$, and because D_{k-1} is a homomorphism, we then know that $D_{k-1}(0) = 0'$ in A'_k. Also, $\partial'_{k+1}(D_k(z_k))$ is some element b'_k of B'_k. Thus we have $f_k(z_k) - g_k(z_k) = b'_k$, so $f_k(z_k) = g_k(z_k) + b'_k$ for some $b'_k \in B'_k$. Consequently

$$\begin{aligned} f_{*k}(z_k + B_k) &= f_k(z_k) + B'_k = (g_k(z_k) + b'_k) + B'_k \\ &= g_k(z_k) + (b'_k + B'_k) = g_k(z_k) + B'_k \\ &= g_{*k}(z_k + B_k), \end{aligned}$$

showing that f_{*k} and g_{*k} are the same homomorphism of $H_k(A)$ into $H_k(A')$.